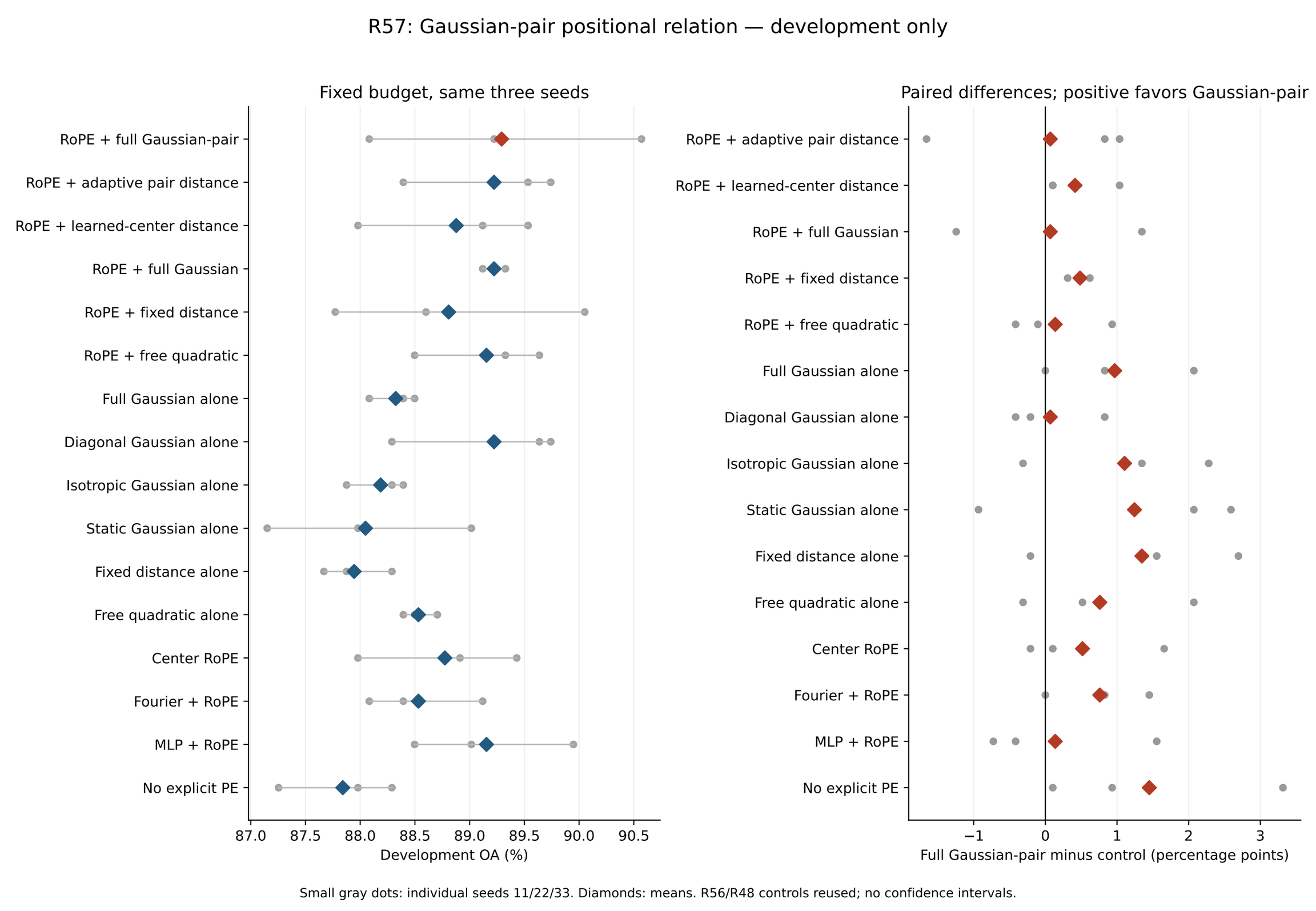
R57: Gaussian-pair positional relation — development only
Fixed budget, same three seeds
RoPE + full Gaussian-pair
RoPE + adaptive pair distance
RoPE + learned-center distance
RoPE + full Gaussian
RoPE + fixed distance
RoPE + free quadratic
Full Gaussian alone
Diagonal Gaussian alone
Isotropic Gaussian alone
Static Gaussian alone
Fixed distance alone
Free quadratic alone
Center RoPE
Fourier + RoPE
MLP + RoPE
No explicit PE
87.0
87.5
88.0
88.5
89.0
89.5
90.0
90.5
Development OA (%)
Paired differences; positive favors Gaussian-pair
RoPE + adaptive pair distance
RoPE + learned-center distance
RoPE + full Gaussian
RoPE + fixed distance
RoPE + free quadratic
Full Gaussian alone
Diagonal Gaussian alone
Isotropic Gaussian alone
Static Gaussian alone
Fixed distance alone
Free quadratic alone
Center RoPE
Fourier + RoPE
MLP + RoPE
No explicit PE
−1
0
1
2
3
Full Gaussian-pair minus control (percentage points)
Small gray dots: individual seeds 11/22/33. Diamonds: means. R56/R48 controls reused; no confidence intervals.